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SOWING DENSITY AND DAMPING-OFF IN PINE SEEDLINGS

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It is generally accepted in forest nursery practice that the risk of damping-off loss in the seed-bed increases in direct proportion to sowing density [1, 2]. This has been demonstrated in the experiments of Hartley [6], but no investigations into the cause of this relationship have been traced.

There are three ways in which increased sowing density might be expected to favour the spread of damping-off pathogens in the seed-bed:—

(*a*) By giving rise to a dense mass of emerged seedlings which would restrict evaporation from the soil and give damp conditions favouring the spread of most damping-off pathogens.

(*b*) By increasing the number of seedlings in the seed-bed which, as a source of nutrient, are available exclusively to the pathogen.

(*c*) By reducing the average distance between individual seedlings, facilitating spread of pathogens from plant to plant, and thereby reducing the competition from soil saprophytes for nutrient in the intervening soil spaces.

The experiments described below have been concerned with damping-off in pine seedlings caused by *Rhizoctonia solani* Kühn. Their object has been to determine whether, in any of these three ways, increases in sowing density bring about increases in damping-off.

Materials and Methods

Experiments were all carried out in the glasshouse at temperatures between 95° F. and 55° F. and relative humidities between 35 per cent and 85 per cent. With the exception of two experiments in which seeds of *Pinus radiata* D. Don. were used, all experiments were made with seeds of *Pinus patula* Schl. & Cham.; these had a germination of 70 per cent to 80 per cent. Seeds were sown in small seed boxes, 7 in. by 8 in. and 4 in. deep, in local forest soil and covered with a uniform quantity (usually 50 gm.) of soil known to contain *R. solani* and a smaller proportion of *Pythium ultimum* Trow. In later experiments a mixture of an 8–10 day-old sand* culture of *R. solani* was sometimes used, mixed with soil. Randomized block designs were employed involving three to five sowing densities replicated six or more times; supplementary treatments were sometimes applied between blocks. Plots were watered sufficiently to keep the soil thoroughly wet but not waterlogged. Disease incidence was recorded a week to ten days after emergence of seedlings; the numbers of healthy and diseased emergent seedlings were counted and, in conjunction with the number of seeds originally sown, percentage values for pre-emergence and post-emergence loss were calculated. In the following tables disease incidence has been stated as post-emergence disease only, as in all experiments the pre- and post-emergence phases of damping-off were closely correlated.

At the end of each experiment in which soil containing pathogens was used as inoculum, diseased plants were sampled from a range of plots, incubated on 2 per cent malt agar at 26° C. for 24 to 36 hours, and examined for pathogens. It was found that plants attacked by *R. solani* and *P. ultimum* could be readily distinguished in this way. Both pathogens have been tested against seedlings of *P. patula* and *P. radiata* from the stage of germination until one week after emergence and have been found to be pathogenic over all this period. This confirms results obtained by Gravatt [5] and Fisher [3].

Results

The results of four experiments to determine the relationship between damping-off and sowing density are shown in Fig 1. The incidence of disease has varied considerably between experiments, but in each the statistical regression obtained between sowing density and damping-off has been highly significant. Pre-emergence and post-emergence (damping-off) loss in each experiment has also been found to be closely correlated (Table I). In all these experiments naturally infected soil was used for inoculum, containing mostly *R. solani* with some *P. ultimum*.

* The medium for these cultures consisted of a mixture of 100 gm. sand, 3 gm. corn meal, and 15 ml. water.

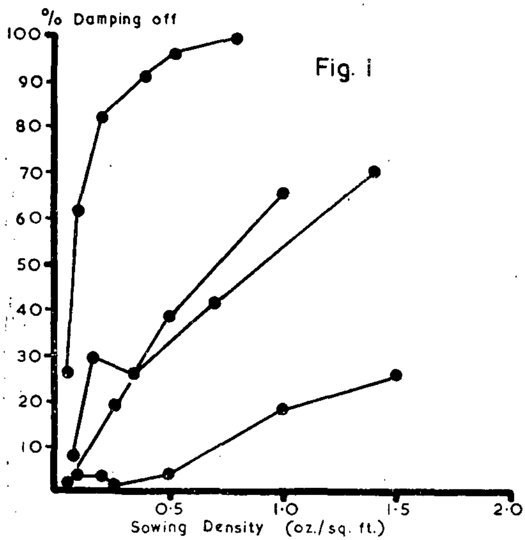


Fig. 1

Soil Moisture

Two experiments have been completed to determine how far variations in soil moisture following increases in sowing density can account for increases in damping-off such as observed above. In both, series of three or four levels of sowing density, replicated six times, were set up. Half the replicate plots were covered with bell jars lined with damp blotting paper, ensuring a saturated atmosphere within, and the other half were left exposed under normal glasshouse conditions. Soil containing a high proportion of *R. solani* to *P. ultimum*

TABLE I.—EFFECTS OF VARIOUS SOWING DENSITIES ON PRE-EMERGENCE AND POST-EMERGENCE LOSS

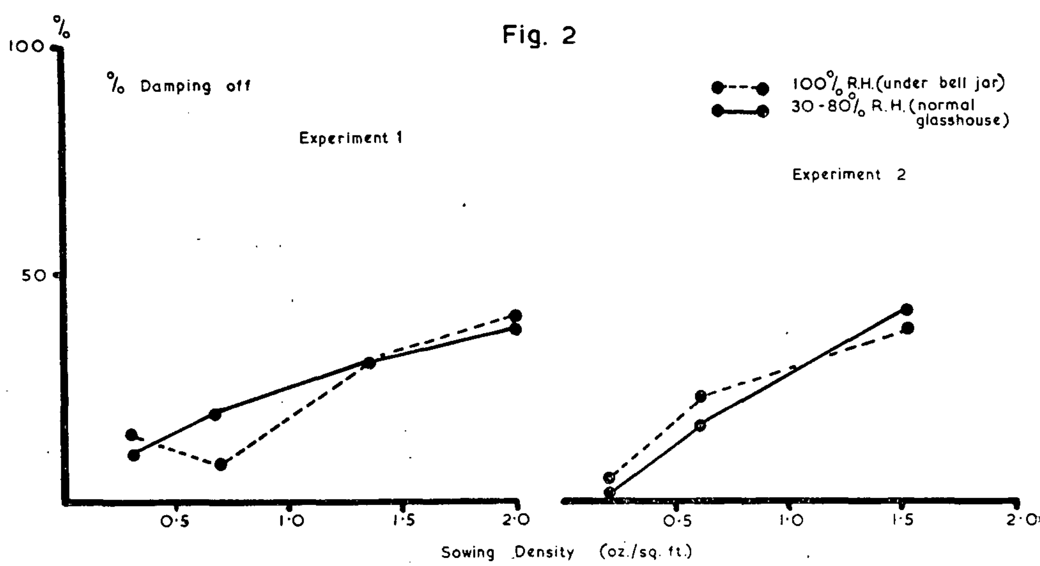
Sowing Density (oz. per sq. ft.)	Damping-off Loss (per cent)			
	Experiment 1		Experiment 2	
	Pre-emergence	Post-emergence	Pre-emergence	Post-emergence
2·10	76·5	66·9	—	—
1·50	—	—	52·8	21·2
1·00	68·9	42·2	41·9	17·8
0·50	41·1	24·3	31·8	4·9
0·25	35·7	24·8	19·7	3·5
0·13	30·7	7·1	—	—

NOTE.—Values for pre-emergence loss include a constant of about 20 per cent due to non-viable seeds in the batch sown.

was used as inoculum. The results of these experiments (Fig. 2) show little difference between the incidence of disease under the two conditions. If increased damping-off at high sowing densities had been the result of restricted evaporation and damp soil conditions, then an equally high disease incidence would have been expected at the lower sowing densities under a saturated atmosphere. As this is not so, it is concluded that variations in soil moisture between high and low sowing densities play little part in causing the corresponding variation of disease incidence.

Further support to this conclusion is found in Table I, where pre-emergence and post-emergence losses are shown to be equally affected by variation in sowing density. As a

Fig. 2



high sowing density can only be expected to modify soil moisture and thus spread of disease in the post-emergence phase, the variation of some other factor with that of sowing density must be postulated to account for the variations of post- and pre-emergence disease observed.

Soil Nutrient

The importance of this factor in determining the relationship between sowing density and damping-off has been investigated in two experiments, in which the following treatments were compared:—

(1) A series of sowing densities without extra treatment.

(2) A similar series to which supplementary quantities of dead seeds were added to each plot to bring the total number of seeds per plot to that of the maximum planted. Supplementary seeds were killed by autoclaving at 15 lb. pressure for five minutes.

A top dressing of soil containing *R. solani* was added uniformly to all plots. Each series was set up in triplicate.

The results of these experiments (Table II) show a consistently higher disease incidence in plots receiving extra dead seeds than in corresponding plots without them. This would be expected if the activity of the pathogen varied with the amount of nutrient available to it. *R. solani* has been described by Garrett [4] as a soil-inhabiting fungus which would be able to colonize the dead seeds added in this experiment and to spread more rapidly as a result. The results shown in Table II show that, within plots sown with mixtures of dead and viable seeds, an increase in incidence of damping-off follows an increase in the proportion of living seeds sown. This may be because live seedlings are more suitable than dead seeds as food bases for the spread of *R. solani,* or due to saprophytic competition to *R. solani* for nutrient in the dead seeds.

The relative effects of nutrient supplied as live and dead seeds on the activity of the pathogen has been investigated in a third experiment in which the following treatments were compared:—

(1) A series of sowing densities without extra treatment.

(2) A similar series to which dead seeds had been added as in the previous experiments.

(3) A series to which supplementary sowings of live seeds, sufficient to bring the total planted per plot to the maximum planted, was added one week after the first sowing. In this way the first sowing emerged and was ready for disease recording before the second sowing had appeared.

TABLE II.—THE EFFECT OF THE ADDITION OF SUPPLEMENTARY DEAD SEEDS TO THE RELATIONSHIP BETWEEN DAMPING-OFF AND SOWING DENSITY

Sowing Density (per sq. ft.)				Damping-off per cent	
Live Seeds		Dead Seeds			
Oz.	No.	Oz.	No.	Experiment 1	Experiment 2
0·18	645	—	—	0·3	1·7
0·36	1290	—	—	5·4	3·9
0·72	2580	—	—	18·6	13·1
0·18	645	0·54	1935	7·1	7·8
0·36	1290	0·36	1290	10·1	8·1
0·54	1935	0·18	645	11·9	—

Each treatment was set up in triplicate. A heavy top dressing of soil known to contain *R. solani* was added to all after the second sowing of live seeds in treatment 3.

The results of this experiment (Table III) were somewhat irregular, but the average disease incidence showed a significant increase from plots without supplementary seeds, through those with extra dead seeds, to those with extra live seeds. This indicates that the living seedling provides a better food base than a dead seed for the spread of *R. solani* in the soil.

TABLE III.—THE EFFECT OF THE ADDITION OF SUPPLEMENTARY LIVING AND DEAD SEEDS ON THE GRADIENT OF DAMPING-OFF WITH SOWING DENSITY

Sowing Density (per sq. ft.)		Damping-off per cent		
Oz.	No.	No extra seeds	Dead seeds added to 0·72 oz. per sq. ft.	Live seeds added to 0·72 oz. per sq. ft.
0·18	645	10·1	18·4	31·9
0·36	1290	16·5	36·6	69·9
0·54	1930	27·2	59·7	46·3
0·72	2580	35·5	—	—
Mean	—	22·3	38·2	49·4

A small laboratory experiment has been carried out to compare the growth rate of *R. solani* in soil and with nutrient derived from germinated seedlings. Sclerotia of *R. solani* were planted singly in tubed 1 per cent soil agar slants; sets of five of these slants were made up with various numbers of germinated pine seeds per tube, killed in the course of sterilization. After intervals of two and four days after inoculation the linear mycelial growth of these cultures was measured. The results (Table IV) show that a significantly greater growth rate after four days, as well as a thicker mycelial mat, was obtained in cultures containing seedlings.

These results not only show that spread of *R. solani* in the soil is closely related to available nutrient, but also that increases in damping-off following increased sowing density are due to corresponding increases in the nutrient available to the pathogen in the soil.

TABLE IV.—THE EFFECT OF THE ADDITION OF GERMINATED SEEDS OF *P. patula* TO SOIL AGAR ON THE GROWTH OF *R. solani*

Linear growth of mycelium (mm.)	Agar Medium			
	Soil only	Soil plus 8 seeds per tube	Soil plus 20 seeds per tube	Soil plus 40 seeds per tube
After 2 days	0·7	0·7	0·8	0·9
After 4 days	0·8	3·3	3·5	4·2

Each treatment replicated five times.

Soil Competition

The effect of competition from other members of the soil microflora on the spread of *R. solani,* and the possibility that the rate of spread of damping-off could be modified through this factor by variations in sowing density, has only been investigated indirectly. Attempts have been made to compare the variation of damping-off within ranges of sowing densities in sterile and normal soils, but these have been unsuccessful due to the rapid reinfestation of sterile soil in the glasshouse.

An approach to this type of experiment has been made by comparing the relationship between damping-off and sowing density in plots of sand or soil. This experiment was sown with a range of six sowing densities, replicated three times each in sand and soil. A 10 per cent mixture of sand culture of *R. solani* in soil was used as inoculum, applied at 50 gm. per plot. Results for disease incidence are given in Fig. 3, and Table V gives

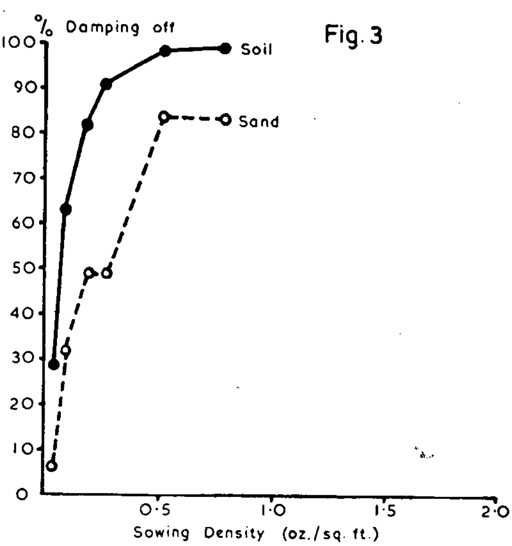


Fig. 3

further information on the chemical and biological status of the sand and soil used. Chemical analysis was performed by Mr. M. T. Friend, and counts for fungi and bacteria were obtained by plating out a 1 : 5000 suspension of soil or sand in sterile saline, using a Czapek-Dox agar medium with additional yeast extract (0.05 per cent).

TABLE V.—ANALYSIS OF SAND AND SOIL AND RESULTS OF PLATING FOR COUNTS OF FUNGI AND BACTERIA

	% Nitrogen	% Carbon	C:N ratio	Total phosphate p.p.m.	pH	Colonies per gram soil	
						Fungi	Bacteria
Soil	0·635	5·43	8·55	105	6·6	80×10^3	110×10^3
Sand	0·005	0·11	22·00	0	7·5	34×10^3	$4{\cdot}5 \times 10^3$

The results show that the greatest incidence of damping-off occurred in the soil with the higher nutrient status and fungal and bacterial population; this would not be expected if competitive rather than nutritional factors modified the rate of spread of the pathogen.

Further evidence has been obtained from a comparison of the relation between sowing density and damping-off in *P. patula* and *P. radiata.* The seedlings of both these species have been found equally susceptible to attack by *R. solani* under comparable conditions in sterile sand. As the seeds of *P. radiata* (33 per gram) are about four times as large as those of *P. patula* (125 per gram), for the same mass of seeds sown per unit area, there will be fewer seeds and a greater distance between seeds of *P. radiata* than those of *P. patula*. If the spread of pathogens from one seedling to another is limited by activity of the soil microflora, then the relationships between sowing density and damping-off for *P. patula* and *P. radiata*, under comparable conditions, will be coincident when plotted on a basis of numbers of seeds per unit area. However, if the nutrient available to the pathogen be a limiting factor, then these relationships would be expected to be coincident when plotted on a basis of mass of seeds per unit area. The results of two experiments to compare the two pine species in these ways are plotted in Fig. 4. Both sets of results show a coincidence of curves for damping-off when sowing density has been plotted as mass of seeds per unit area, which would be expected if the spread of the pathogen was limited by available nutrient rather than competition from other members of the soil microflora.

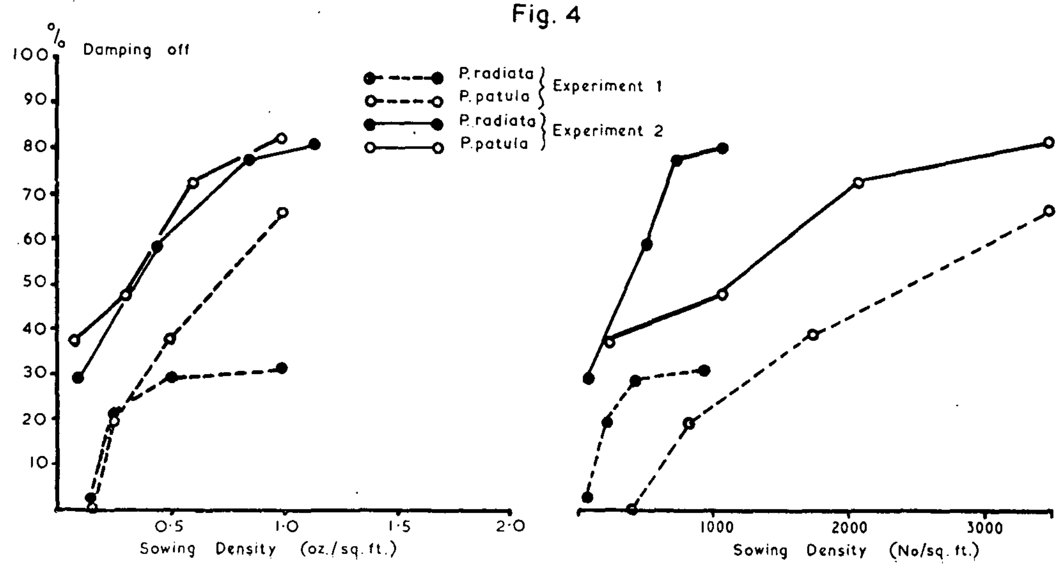
A third experiment has also given results showing that the spread of *R. solani* as a damping-off pathogen is largely dependent on available nutrient. In this the disease incidence over a range of four sowing densities was compared after various mixtures (10, 40 and 80 per cent) of a sand culture of *R. solani* and soil had been applied as inoculum (50 gm. per plot). The results (Table VI) show that where inoculum has been added to plots, a marked gradient of disease with sowing density has been obtained, irrespective of the constitution of the inoculum. The mean values for disease incidence at the three inoculum levels fall within a small range of each other.

TABLE VI.—THE EFFECT OF VARYING QUANTITIES OF INOCULUM OF *R. solani* ON THE RELATIONSHIP BETWEEN DAMPING-OFF AND SOWING DENSITY OF *P. patula*

Sowing Density per sq. ft.		Damping-off (per cent)				
		Per cent composition of added inoculum				Mean
Oz.	No.	0	10	40	80	
0·041	180	0	4·5	12·3	25·9	10·7
0·220	771	21·5	58·0	68·9	53·6	50·5
0·366	1285	33·4	77·3	73·1	75·6	64·8
0·686	2056	6·9	80·4	92·8	86·6	66·7
Mean ..	—	15·4	55·1	61·8	60·4	—

If competition from other members of the soil microflora influenced the rate of spread of the pathogen, then this effect would be overwhelmed by graded increases in inoculum with accompanying increases in mean disease incidence. Similarly, the gradient of disease incidence with sowing density would disappear at high inoculum levels. On the other hand, if nutrient as seeds and seedlings is a major factor in determining the rate of spread of the pathogen, then a gradient of disease with sowing density would be expected irrespective of inoculum level and the mean disease incidence at all inoculum levels would be expected

Fig. 4



to be about the same. On this basis the results show that available nutrient is clearly the major factor in determining activity of the pathogen in relation to sowing density.

CONCLUSION

Throughout these experiments results have consistently shown that the activity of *R. solani* as a damping-off pathogen is largely dependent on the nutrient available in the soil. Furthermore, increased damping-off losses regularly associated with high rates of sowing have been shown to be due to accompanying increases of nutrient in the soil rather than to any modification of soil moisture conditions or to changes in the degree of competition to the spread of the pathogen through the soil.

The demonstration that dead seeds in the soil may act as nutrient sources for *R. solani* and lead to an increased spread of the pathogen has a practical application. Seeds of low germination are sometimes thickly sown in the bed to obtain reasonable emergent populations. Providing the dead seeds in such defective batches are not empty shells, this practice will lead to an increase in soil nutrient and an increase in the risk of damping-off from *R. solani*.

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